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import numpy as np
import matplotlib.pyplot as plt

# Define the sigmoid function
def sigmoid(z):
    return 1 / (1 + np.exp(-z))

# Generate x values
x = np.arange(-5, 5, 0.1)

# Plot the sigmoid curve
plt.figure(figsize=(8, 5))
plt.plot(x, sigmoid(x), color='pink', linewidth=2)

plt.title('Visualization of the Sigmoid Function')
plt.xlabel('Input (z)')
plt.ylabel('Sigmoid(z)')
plt.grid(True)

plt.show()
```

